



Research article

Waveform Design, Not Amplitude, Governs Safe Memory Discrimination in a Strong-Allee Predator–Prey Model

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Abstract: A fitted fractional order α does not identify a fractional mechanism: on a finite horizon $[0, T]$, delayed feedback or a few hidden states reproduce similar trajectories. We ask when a *safe* controlled experiment can separate these mechanisms in a strong-Allee predator–prey system with coexistence equilibrium $x^* = 2/3$, $y^*(A) = 2(2 - 3A)/(9A)$, whose model, locked parameters and Jacobian $J(A)$ are inherited from a previously submitted companion paper and are not claimed as new. Two obstructions are separated. The first is structural and is *not* binding: the collocated Caputo transfer decays as $O(\omega^{-\alpha})$, which no finite rational latent transfer and no strictly proper retarded-delay model reproduces exactly. The second is quantitative and *is* binding: positive exponential mixtures approximate the memory kernel in $L^1(0, T)$ at a root-exponential rate, so that $m(\varepsilon) = O(\log^2(1/\varepsilon))$ hidden states suffice; computed directly at the prey-response level, the L^1 surrogate error falls from 1.29 at $m = 4$ to $4.8 \cdot 10^{-5}$ at $m = 128$ and the induced minimax testing error rises to 0.49999, so at attainable latent orders the finite-state rival is indistinguishable. Against that obstruction we test whether safe excitation must sacrifice information. It must not. In a four-class BIC benchmark over three amplitudes (2268 cells, no divergences) reducing amplitude fails as a safety strategy — PRBS still crosses the Allee threshold in 71% of cells at half amplitude while remaining the most informative design — and only one of six classical waveform families is safe, scoring macro-accuracy 0.514 against 0.537 overall. Searching 163 840 candidate waveforms outside those families under a hard margin constraint at the *same* amplitude budget yields piecewise-constant designs that are both safe and far more informative: macro-accuracy **0.803** with zero threshold crossings, against 0.514 for the only safe classical design and 0.763 for the best classical design of any kind, which crosses in 35.7% of cells. A posteriori validated integration certifies these trajectories, whereas an invariant-ball certificate admits only $\|u\|_\infty \leq 1.4 \cdot 10^{-3}$, 70× smaller. The safety–informativeness trade-off reported for classical excitation families is therefore an artifact of the waveform parameterisation, not a property of the system; what survives is the latent-complexity obstruction, since even the best safe design recovers only 0.475 recall on the finite-latent rival. Safe identification of ecological memory is limited by the declared rival-complexity budget m , not by the safety margin.

Keywords: fractional-order dynamics; time-delay systems; model discrimination; experimental design; Allee effect; predator–prey dynamics

Mathematics Subject Classification: 34A08; 34K20; 62K05; 92D25

1. Introduction

Caputo derivatives of order $0 < \alpha < 1$ model hereditary effects in viscoelasticity, anomalous transport, and population dynamics [4, 6, 11, 16]; fractional predator–prey models in particular form a sizeable literature [1, 10, 17, 18]. An

empirical difficulty sits underneath all of it. Estimating $\alpha < 1$ from data does not establish a fractional mechanism, because on a bounded window $[0, T]$ a delay term $\xi(t - \tau)$ or a low-dimensional latent state $q \in \mathbb{R}^m$ generates trajectories within the noise floor of the fractional ones (Theorem 3.2 makes this quantitative). Design methods for fractional-order *estimation* exist [8, 9, 13, 20]; safety-constrained information acquisition has been analyzed in general settings [15, 19]. The intersection — discrimination between memory *mechanisms*, under an ecological safety constraint — is the subject here:

Which perturbations and measurements can distinguish fractional memory from delayed or latent alternatives in a strong-Allee ecological system, and when does ecological safety make that discrimination fundamentally difficult?

Safety is not decorative in this question. Below the Allee threshold A the prey growth term $P(x) = rx(1 - x/K)(x/A - 1)$ turns negative and collapse is irreversible, so the perturbations most informative about memory ($\|u\|_\infty$ near the budget, broadband spectra) are the ones most likely to destroy the object of study. Section 8 measures this numerically: the best-discriminating input crosses the threshold in 100% of stable-regime cells.

The ecological backbone is not built here. The equations, the locked parameters $(r, K, a, h, e, m) = (\frac{3}{2}, 1, 1, \frac{1}{2}, \frac{4}{5}, \frac{2}{5})$, the equilibrium $x^* = 2/3$, $y^*(A) = 2(2 - 3A)/(9A)$, and the Jacobian $J(A)$ all come from a companion study [3, 5, 12, 21] where they were certified by validated numerics. Reuse is deliberate: with the ecology frozen, the discrimination question can be isolated from any change in biological mechanism. Nothing from [12] is claimed as new. The new objects are the four rival response laws (Section 2), the perturbation/observation architecture, the finite-horizon discrimination bounds of Section 3, the combined fractional-delay model (2.4), and the benchmark of Section 8. Only the ODE and Caputo mechanisms share $J(A)$ exactly; the delay and latent rivals are calibrated alternative response laws matched at the same operating point, input channel, and units.

Explicit relationship to the previously submitted paper. The two manuscripts form a sequence, so we state the reuse boundary rather than leave it implicit. Exactly four ingredients are inherited from [12]: the strong-Allee equations, the locked parameters, the coexistence equilibrium, and $J(A)$. None re-enters the novelty claim; they define the operating and safety geometry on which a different problem is posed. Table 1 gives the division of labour, axis by axis.

Table 1. Relationship between the previously submitted companion paper and the present manuscript. The ecological backbone is intentionally shared; the research questions, analytical targets, simulations, and headline results are different.

Aspect	Previously submitted paper [12]	Present paper
Scientific question	Certify equilibria, stability, the Matignon boundary, extinction/boundedness, and discretization-induced stability inversion.	Design and evaluate experiments that distinguish Caputo, delayed, integer-order, and finite-latent memory mechanisms.
Shared material	Strong-Allee ecological equations, locked parameters, coexistence equilibrium, and Jacobian.	The same ecological backbone is reused only as the operating and safety test bed.
Primary mathematics	Validated numerics, Krawczyk/interval certificates, Matignon classification, global extinction/boundedness statements.	Structural transfer separation, finite-horizon latent approximation, testing obstruction, and finite-state prey-response approximation.
Primary computation	Certified parameter grid and certificate inventory.	Frequency-response diagnostics, input-design experiments, four-class model-discrimination benchmark, and fractional-delay simulations.
Headline output	Certified dynamical regimes of the Caputo ecological model.	Safety-informativeness limits for active discrimination of competing memory mechanisms.

The methodological separation matters as well. A stability certificate decides whether a specified Caputo model is consistent with a dynamical regime; it does not decide whether that model can be told apart from a delay or latent mechanism reproducing the same observations to within σ . Conversely, no result below recertifies the backbone. We condition on the certified operating geometry of [12] and ask how much perturbation and measurement information the discrimination task needs on top of it.

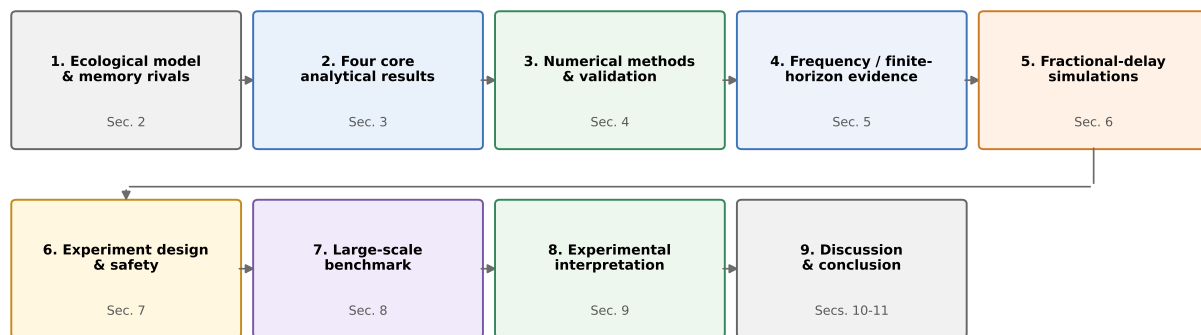
Main contributions. Four contributions organize the paper.

1. **Structural and finite-horizon separation** (Theorems 3.1 and 3.2). The collocated Caputo transfer decays as $(CB)\tau_0^{1-\alpha}s^{-\alpha} + O(|s|^{-2\alpha})$, a noninteger signature unreachable by finite rational latent transfers or by the declared strictly proper retarded class ($O(\omega^{-1})$ decay). Simultaneously, positive exponential mixtures with m modes reach any prescribed $L^1(0, T)$ tolerance, so separation is meaningful only relative to a declared budget m .

2. **Certified prey-response testing obstruction** (Theorems 3.3 and 3.4). On the linearized prey channel, a pole/branch decomposition yields finite-state surrogates with interval-enclosed L^1 errors (0.5335 at $m = 4$ down to 0.0070 at $m = 32$; $\alpha = 0.85$, $A = 0.25$, $T = 12$). Under the pulse budget $\|u\|_2 \leq 0.120$ and $\sigma = 0.10$, the Pinsker minimax bound rises from 0.340 to 0.498 — to chance, effectively.
3. **Fractional-delay stress test** (Section 6). The combined model (2.4) equals the pure Caputo system at $\tau = 0$ and the DDE at $\alpha = 1$; simulations over $(\alpha, \tau) \in [0.70, 1] \times [0, 0.5]$ show delay shifting the predator peak by up to ≈ 0.7 time units, eroding the minimum Allee margin to 0.238 at $\alpha = 0.95$, and generating interior responses far from both limits (Table 7).
4. **Safety–informativeness trade-off** (Tables 11 and 8). Across six input families: PRBS and multisine discriminate best (0.584, 0.583 macro-accuracy) with Allee-crossing rates of 1.000 and 0.357; multiscale never crosses and scores 0.324. The constrained design problem is measured, not merely postulated.

Complete proofs, secondary results, the Bayesian sequential extension, and solver derivations are in the Supplementary Material; the main text keeps the four statements needed to read the numerics.

Paper architecture: compact theory, expanded simulations, and a dedicated fractional-delay study



Theory is compacted in the main text; detailed proofs and the Bayesian extension are in the Supplement.

Figure 1. Structure of the paper. Sections 2–3: ecological model, rival memory laws, and the four analytical results. Sections 4–5: solver validation (Table 3) and frequency/finite-horizon evidence. Section 6: fractional-delay simulations. Sections 7–8: input design, safety, and the 1512-cell benchmark. Proofs and secondary theory: Supplementary Material.

2. Ecological Model and Competing Memory Mechanisms

2.1. Strong-Allee predator–prey backbone

All objects in this subsection are inherited from [12], where they were derived and certified with validated numerics: the model, the locked parameters, the equilibrium $(x^*, y^*(A))$, the Jacobian $J(A)$, and the Matignon boundary $\alpha^*(A)$. They are restated because the experiments of Sections 4–8 are defined in terms of them; none is claimed as new.

The nonlinear model is

$$f_1(x, y) = rx \left(1 - \frac{x}{K}\right) \left(\frac{x}{A} - 1\right) - \frac{axy}{1 + hx}, \quad (2.1)$$

$$f_2(x, y) = e \frac{axy}{1 + hx} - my, \quad (2.2)$$

with locked parameters

$$r = \frac{3}{2}, \quad K = 1, \quad a = 1, \quad h = \frac{1}{2}, \quad e = \frac{4}{5}, \quad m = \frac{2}{5}.$$

The coexistence equilibrium and Jacobian are

$$x^* = \frac{2}{3}, \quad y^*(A) = \frac{2(2 - 3A)}{9A},$$

$$J(A) = \begin{pmatrix} \frac{7A-2}{10A} & -\frac{1}{2} \\ \frac{8A}{2-3A} & 0 \end{pmatrix}.$$

For the commensurate Caputo model, local stability follows Matignon's sector condition $|\arg \mu| > \alpha\pi/2$ on the eigenvalues μ of $J(A)$ [14]. In the complex-pair regime $2/7 < A < 10/19$ the critical order is

$$\alpha^*(A) = \frac{2}{\pi} \arctan\left(\frac{\sqrt{4D(A) - T(A)^2}}{T(A)}\right),$$

where $T(A) = \text{tr } J$ and $D(A) = \det J$. At $A = 0.30$, $\alpha^* \approx 0.969012276$: the Caputo system with $\alpha = 0.9 < \alpha^*$ is stable while its integer-order limit ($\alpha = 1 > \alpha^*$) is unstable [12]. Most discrimination experiments below use $A \leq 2/7$, where every candidate mechanism is stable at the operating point and stability cannot serve as a discrimination shortcut.

Inherited backbone versus new experimental layer. Notation is kept identical to [12] for an identification reason: if the ecological equations changed between papers, a difference in simulated observations could reflect the altered biology instead of the memory law. The ecology is therefore frozen and only the response law varies — ${}^C D_t^\alpha$, the delay $\xi(t - \tau)$, or a latent state $q \in \mathbb{R}^m$ — with all candidates facing the same budget $\|u\|_\infty \leq 0.10$, horizon $T = 12$, and threshold A . Table 2 lists the frozen quantities.

Table 2. Locked ecological backbone used by both manuscripts. These values are inherited inputs, not new contributions of the present paper.

Quantity	Value	Role in present experiments
r	1.5	intrinsic prey growth
K	1	prey carrying capacity
a	1	attack-rate scale
h	0.5	Holling-II handling parameter
e	0.8	conversion efficiency
m	0.4	predator mortality
x^*	$2/3$	common coexistence prey level
$y^*(A)$	$2(2 - 3A)/(9A)$	coexistence predator level
$J(A)$	Eq. above	ODE/Caputo local ecological generator

The threshold A has two roles. First, $x(t) < A$ defines the failure event: $P(x) < 0$ there, and the trajectory can leave the coexistence basin permanently (Figure 3, left). Second, $x^* - A$ is the natural safety margin; at $A = 0.25$ it equals $2/3 - 0.25 \approx 0.417$. An input can thus score highly on the information criterion of Section 7 and still be ecologically inadmissible — Tables 11 and 8 quantify exactly this.

2.2. Four rival mechanisms

With the reference time τ_0 enforcing dimensional consistency [6, 11, 16], the four candidate mechanisms around the operating point are

$$\begin{aligned} \text{ODE: } \dot{\xi} &= J\xi + Bu, \\ \text{Caputo: } \tau_0^{\alpha-1} {}^C D_t^\alpha \xi &= J\xi + Bu, \\ \text{DDE: } \dot{\xi} &= A_0 \xi(t) + A_1 \xi(t - \tau) + Bu, \\ \text{latent: } \dot{q} &= \bar{A}q + \bar{B}u, \quad y = \bar{C}q. \end{aligned}$$

The Caputo mechanism is the only one carrying a fractional order. The latent class, which would otherwise be flexible enough to imitate anything, is held to an explicit state-dimension budget m — a restriction whose necessity is precisely the content of Result II below.

System, observation channels, and rival memory laws

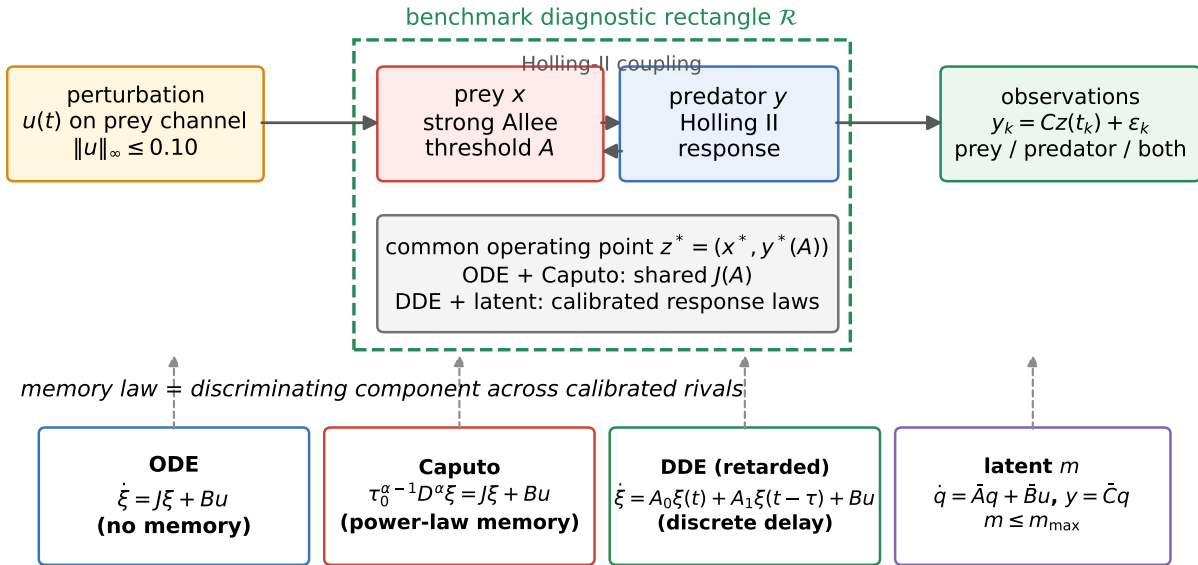


Figure 2. Setting: backbone from [12]; perturbation $u(t)$ on the prey channel with $\|u\|_\infty \leq 0.10$; observations $y_k = Cz(t_k) + \varepsilon_k$ on prey, predator, or both. ODE and Caputo share $J(A)$ exactly; DDE and latent are calibrated rival response laws. Dashed rectangle: numerical benchmark diagnostic region only (not an invariant-set certificate).

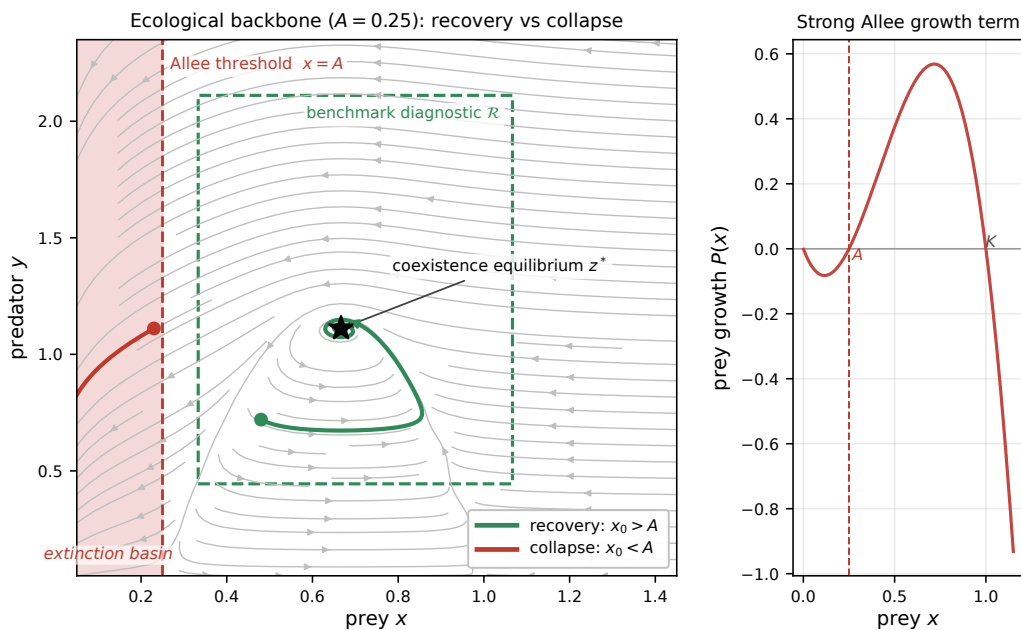


Figure 3. Safety geometry at $A = 0.25$ (backbone from [12]). Left: recovery for $x_0 > A$ versus collapse for $x_0 < A$. Right: $P(x) = rx(1 - x/K)(x/A - 1)$, with sign change at $x = A$.

Combined fractional-delay stress-test model. A system may carry hereditary memory and a trophic lag simultaneously, so we also simulate the combined model

$$\tau_0^{\alpha-1} D_t^\alpha x(t) = P(x(t); A) - \frac{ax(t)y(t)}{1 + hx(t)} + u(t), \quad (2.3)$$

$$\tau_0^{\alpha-1} D_t^\alpha y(t) = e^{\frac{ax(t-\tau)y(t)}{1 + hx(t-\tau)}} - my(t), \quad (2.4)$$

with constant equilibrium history on $[-\tau, 0]$. Its boundaries are exact: at $\tau = 0$, (2.4) is the pure Caputo model; at $\alpha = 1$ it is the retarded DDE benchmark. Both limits are verified numerically to 6.75×10^{-6} and better in Table 3. The model is a numerical stress test only; no theorem depends on it.

3. Analytical Backbone: Four Results Used by the Experiments

The numerical study rests on four analytical statements. Complete proofs, the secondary theorems (optimal input design, multisine support, safety invariance, the sequential information identity), and the Bayesian layer are in the Supplementary Material; here each statement appears with the minimum context needed to use it in Sections 5–8.

3.1. Result I: structural separation

On the collocated channel ($CB \neq 0$) the Caputo transfer is

$$G_\alpha(s) = C(q_\alpha(s)I - J)^{-1}B, \quad q_\alpha(s) = \tau_0^{-1}(s\tau_0)^\alpha,$$

with high-frequency expansion

$$G_\alpha(s) = (CB)\tau_0^{1-\alpha}s^{-\alpha} + O(|s|^{-2\alpha}).$$

Theorem 3.1 (Structural separation). *For $0 < \alpha < 1$ and $CB \neq 0$, the Caputo transfer cannot equal any finite-dimensional rational latent transfer on a nonempty open subset of a common analytic domain. It also cannot equal a strictly proper finite-state retarded model with finitely many discrete delays, because the latter has $O(\omega^{-1})$ high-frequency decay whereas the Caputo channel has $O(\omega^{-\alpha})$ decay.*

The $O(\omega^{-\alpha})$ signature is thus exactly unreachable for both declared rival classes. The statement concerns exact equality of transfers on open frequency sets; it makes no promise about experiments with finite bandwidth, horizon $T < \infty$, and noise $\sigma > 0$. Theorem 3.2 covers that regime.

3.2. Result II: finite-horizon latent approximation

The Caputo kernel $k_\alpha(t) = t^{\alpha-1}/\Gamma(\alpha)$ admits a diffusive (spectral-mixture) representation; on $[0, T]$ a constructive positive exponential sum $k_m(t) = \sum_{j=1}^m c_j e^{-\lambda_j t}$, $c_j \geq 0$, achieves

$$\|k_\alpha - k_m\|_{L^1(0,T)} \leq \widehat{E}_m, \quad \widehat{E}_m \downarrow 0$$

along an expanding nested rate hierarchy.

Theorem 3.2 (Finite-horizon approximation and complexity barrier). *For every $T < \infty$ and $\varepsilon > 0$, there exists a finite positive exponential mixture whose $L^1(0, T)$ distance from the Caputo kernel is below ε . Consequently, without a finite complexity restriction on the latent rival, no fixed finite-horizon experiment can guarantee a uniform positive separation from the entire latent hierarchy.*

Results I and II fix the problem geometry: exact separation (Theorem 3.1) coexists with ε -approximation at finite T (Theorem 3.2). Discrimination is therefore relative to a declared budget m ; in the benchmark of Section 8 the latent class is capped at $m = 3$, and in the certified computation of Theorem 3.4 the budget runs over $m \in \{4, 8, 16, 32\}$.

3.3. Result III: testing obstruction under a physical input budget

Let S denote the observation operator, $R > 0$ the Gaussian noise covariance, and

$$C_{\text{obs}} = \frac{1}{2} \|R^{-1/2}S\|^2.$$

Young's convolution inequality maps the L^1 kernel error into an L^2 observation-space bound, $\|(k_\alpha - k_m) * u\|_2 \leq \widehat{E}_m \|u\|_2$; a two-point Gaussian argument converts that bound into a minimax statement.

Theorem 3.3 (Complexity-dependent testing obstruction). *If the admissible experiment class satisfies $\|u\|_2 \leq B$, then the composite minimax error for testing the fractional response against a latent hierarchy containing an explicit competitor with error $\widehat{E}_m$ obeys*

$$P_e^*(m) \geq \Psi\left(C_{\text{obs}} \widehat{E}_m^2 B^2\right),$$

where Ψ is any valid equal-covariance Gaussian two-point lower bound (Pinsker is used in the numerical plots). Thus $P_e^*(m) \rightarrow 1/2$ whenever $\widehat{E}_m \rightarrow 0$ and B remains bounded.

Since $\widehat{E}_m \downarrow 0$ while B and C_{obs} stay fixed, $P_e^*(m) \rightarrow 1/2$: under a capped input budget, a sufficiently rich rival pushes every test toward the equal-prior chance level, independently of the experiment design.

3.4. Result IV: ecological prey-response approximation

Results I–III concern the abstract kernel k_α . Result IV addresses the measured object itself, the prey-to-prey transfer g_α of the linearization at $(x^*, y^*(A))$. The factorization $(s^\alpha I - J)^{-1} = (sI - J)^{-1} s^{-\alpha}$, though tempting, is false and is not used. Instead g_α is split by contour deformation into its principal-sheet pole pair — realized *exactly* by a stable two-state integer-order block — plus a branch-cut remainder, smooth on $(0, \infty)$, which is the only part requiring approximation.

Theorem 3.4 (Certified finite-state approximation of the prey response). *For the displayed Matignon-stable cells and $m \in \{4, 8, 16, 32\}$, there exists an explicit stable finite-state prey-response surrogate g_m such that*

$$\|g_\alpha - g_m\|_{L^1(0,T)} \leq \widehat{E}_m^{\text{state}},$$

where the bound is obtained by outward-rounded interval subdivision and a rigorous tail estimate. At $\alpha = 0.85$, $T = 12$, the certified enclosure at $A = 0.25$ decreases from 0.5335 ($m = 4$) to 0.0070 ($m = 32$).

Every displayed constant is an outward-rounded interval enclosure, not a floating-point estimate. The declared rectangular pulse has support length 1.44, so $\|u\|_\infty \leq 0.10$ implies $\|u\|_2 \leq 0.10\sqrt{1.44} = 0.120$; combining with $\sigma = 0.10$ in Theorem 3.3 gives Pinsker bounds 0.340, 0.462, 0.492, 0.498 at $A = 0.25$ for $m = 4, 8, 16, 32$ (Table 5 adds $A = 0.30$). At $m = 32$ the gap to the chance level 0.500 is 0.002: the surrogate is statistically indistinguishable from the fractional response under this protocol.

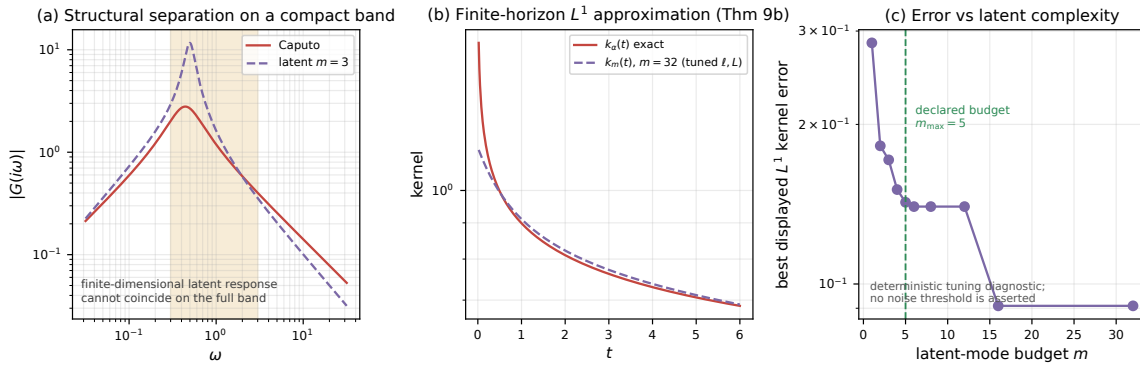


Figure 4. Compatibility of Theorems 3.1 and 3.2: the Caputo transfer keeps its noninteger signature (exact separation), while m -mode exponential mixtures reduce the $L^1(0, T)$ kernel error monotonically in m (finite-horizon approximation).

4. Numerical Methods and Validation

The computational layer supports the analytical claims of Section 3; it does not replace them. All scripts and the machine-readable outputs (JSON/NPZ) behind every figure are released with the paper, so each displayed number can be regenerated from source.

4.1. Solvers

ODE and latent-state systems are integrated by classical fourth-order Runge–Kutta with step $h = T/N$; retarded DDEs by an interpolated-history RK4 variant. Caputo systems use the fractional Adams–Bashforth–Moulton PECE

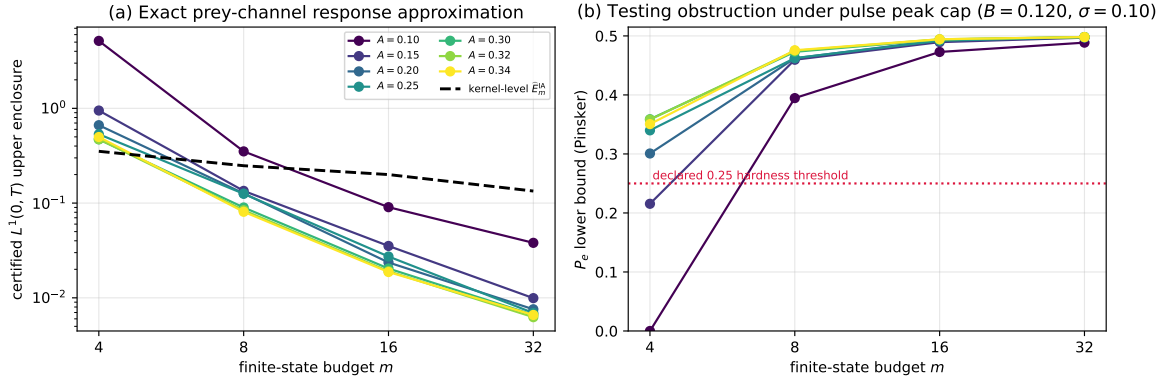


Figure 5. Theorem 3.4 in numbers. Left: interval-enclosed $\widehat{E}_m^{\text{state}}$ versus m at $\alpha = 0.85$, $T = 12$. Right: induced Pinsker lower bound under the pulse cap $\|u\|_2 \leq 0.120$, $\sigma = 0.10$, approaching the chance level $1/2$ as m grows.

scheme [16]; its scalar implementation is validated against the exact Mittag–Leffler solution $E_\alpha(\lambda t^\alpha)$, with maximum errors 8.70×10^{-4} , 4.64×10^{-4} , 2.52×10^{-4} at $N = 250, 500, 1000$ (Table 3).

The combined model (2.4) reuses the PECE history convolution and evaluates the delayed prey value $x(t - \tau)$ by linear interpolation of the already computed trajectory. Reported grids with $\tau > 0$ satisfy $\tau \geq h = T/N$, so the corrector never requires unknown history. The model’s two boundaries serve as validation gates:

- at $\tau = 0$ the code delegates to the validated Caputo solver (no duplicated computation);
- at $\alpha = 1$ the output is compared against the independent DDE RK4 solver.

At $A = 0.25$, $\tau = 0.35$, the $\alpha = 1$ discrepancy against the DDE solver is 6.75×10^{-6} ($N = 200$), 1.68×10^{-6} ($N = 400$), 4.20×10^{-7} ($N = 800$); the error falls by a factor of ≈ 4 per grid doubling. In the interior, at $(\alpha, \tau) = (0.85, 0.35)$, self-convergence against the $N = 800$ reference improves from 2.18×10^{-3} ($N = 200$) to 8.05×10^{-4} ($N = 400$).

4.2. Inputs, observations, and metrics

Six bounded input families are compared under the common peak budget $\|u\|_\infty \leq 0.10$: pulse, multiscale pulse train, sinusoid, multisine, chirp, and pseudorandom binary sequence (PRBS); see Figure 13. Observation channels: prey, predator, or both. Numerical safety is reported separately from solver survival via the minimum Allee margin

$$\min_t (x(t) - A)$$

and the fraction of runs crossing $x = A$.

Four additional diagnostics apply to the fractional-delay runs: the L^2 trajectory distance to the pure Caputo limit ($\tau = 0$), the distance to the pure DDE limit ($\alpha = 1$), the predator peak time and amplitude, and the bridge score $\min(d_{\text{Caputo}}, d_{\text{DDE}})$, which is small whenever *either* pure mechanism explains the data and large only when neither does.

4.3. Validation matrix and numerical resolution

Each numerical claim is tied to an explicit convergence gate rather than to a single solver run; Table 3 collects all eight. Three reference types appear: the exact Mittag–Leffler solution (Caputo gates), the independent DDE RK4 solver ($\alpha = 1$ boundary), and an $N = 800$ self-convergence reference (interior cell). All eight error sequences decrease monotonically in N .

The gates are deliberately heterogeneous. Two implementations sharing one discretization can agree while both are wrong; the informative comparisons are against mathematically different references. Hence the choices above: delegation (not duplication) at $\tau = 0$, and an independent integer-order solver at $\alpha = 1$, where the memory convolution degenerates. The two checks constrain opposite boundaries of (2.4).

4.4. Experimental grid and reproducibility

The benchmark uses a small set of interpretable axes (Table 4), all under the actuator budget $\|u\|_\infty \leq 0.10$. The four-class stable benchmark sweeps $A \in \{0.20, 0.25\}$, $\alpha \in \{0.70, 0.85, 0.90, 0.95\}$, three SNR levels, three observation

Table 3. Numerical validation gates. The Caputo column uses the exact Mittag–Leffler reference; the fractional-delay boundary uses an independent DDE RK4 solution.

Gate	grid size	Reference	max. abs. discrepancy
Caputo PECE	$N = 250$	exact Mittag–Leffler	8.70×10^{-4}
Caputo PECE	$N = 500$	exact Mittag–Leffler	4.64×10^{-4}
Caputo PECE	$N = 1000$	exact Mittag–Leffler	2.52×10^{-4}
Fractional-delay, $\alpha = 1$	$N = 200$	DDE RK4	6.75×10^{-6}
Fractional-delay, $\alpha = 1$	$N = 400$	DDE RK4	1.68×10^{-6}
Fractional-delay, $\alpha = 1$	$N = 800$	DDE RK4	4.20×10^{-7}
Fractional-delay, $(\alpha, \tau) = (0.85, 0.35)$	$N = 200$	$N = 800$	2.18×10^{-3}
Fractional-delay, $(\alpha, \tau) = (0.85, 0.35)$	$N = 400$	$N = 800$	8.05×10^{-4}

channels, and six waveforms, at 200 replicates per cell (1512 cells; Section 8). The fractional-delay campaign is intentionally smaller: $A = 0.25$, $\alpha \in \{0.75, 0.85, 0.95\}$, $\tau \in \{0.15, 0.30, 0.45\}$ for the design comparison, and the 7×6 grid $\alpha \in \{0.70, \dots, 1.00\}$, $\tau \in \{0, \dots, 0.50\}$ for the bridge maps.

Table 4. Main numerical-design axes. The fractional-delay campaign is intentionally targeted and is separate from the larger four-class benchmark.

Component	Values / protocol
Peak input budget	$\ u\ _\infty \leq 0.10$ for all six waveform families
Four-class stable benchmark	ODE, Caputo, retarded DDE, latent-3; $A \in \{0.20, 0.25\}$; 200 replicates per cell
Fractional orders in benchmark	$\alpha \in \{0.70, 0.85, 0.90, 0.95\}$ where applicable
Observation channels	prey only, predator only, both species
Noise regimes	high, medium, and low SNR
Fractional-delay design grid	$A = 0.25$, $\alpha \in \{0.75, 0.85, 0.95\}$, $\tau \in \{0.15, 0.30, 0.45\}$
Fractional-delay bridge grid	$\alpha = 0.70, 0.75, \dots, 1.00$ and $\tau = 0, 0.10, \dots, 0.50$
Simulation horizon	$T = 12$ for the principal trajectory/design comparisons

Every figure quantity traces to a JSON or NPZ file written by the simulation scripts; the fractional-delay script emits its validation sequence, design ranking, and bridge arrays together with Figures 9–12. Separating simulation output from rendering allows plots to be redrawn without rerunning the archived 1512-cell benchmark, and the archived statistics to be audited without redrawing.

5. Frequency-Domain and Finite-Horizon Evidence

The structural results of Section 3 are visible in two frequency-domain plots, computed before any benchmark. Figure 6 compares $|G(i\omega)|$ on the prey channel at $A = 0.25$ for the four mechanisms; Figure 7 shows the phase, which on the collocated channel converges to $-\alpha\pi/2 \approx -76.5^\circ$ for $\alpha = 0.85$ — the scale-free fractional signature of Theorem 3.1.

5.1. From exact signatures to finite-horizon ambiguity

The two panels must be read jointly with Theorem 3.2. The $O(\omega^{-\alpha})$ decay and the $-\alpha\pi/2$ phase identify the Caputo channel asymptotically, but an experiment observes a finite band $[\omega_{\min}, \omega_{\max}]$, a finite window $[0, T]$, and noise of level σ . On that restricted set a latent model with enough modes matches the observable part of the response while remaining analytically distinct — the coexistence of exact separation (Theorem 3.1) and finite-horizon mimicry (Theorem 3.2) stated in frequency-domain terms.

The prey-response computation restates the point in the instrumented channel. Table 5 lists the certified L^1 bounds $\widehat{E}_m^{\text{state}}$ and the induced testing floors at $\alpha = 0.85$, $T = 12$, $\sigma = 0.10$: from $m = 4$ to $m = 32$ the enclosure shrinks by a factor of 77 (from 0.5334 to 0.0069 at $A = 0.25$) and the testing bound rises from 0.374 to 0.498, i.e., to within 0.002 of the equal-prior chance level. Two bounds are reported. Since the two hypotheses are Gaussian with common covariance $\sigma^2 I$, the exact minimax error $\Phi(-d/2)$ with $d = \|\Delta\mu\|_2/\sigma$ is available and is uniformly tighter than the Pinsker bound; the difference matters only at small m , where Pinsker understates the floor by 0.034 at $m = 4$. We quote the exact bound and retain Pinsker for comparability with the literature.

Because Table 5 stops at $m = 32$ it shows only part of the obstruction. Table 6 continues it to $m = 128$ using an explicit positive-weight exponential-sum surrogate constructed as in Lemma 3.2 rather than the interval-optimised class $\mathcal{G}_m$. Because that surrogate is one admissible element and not the infimum, its error exceeds $\widehat{E}_m^{\text{state}}$ — by factors 2.4 at $m = 4$ and 3.3 at $m = 32$ where the two overlap — so the induced floors are valid and conservative. They reach

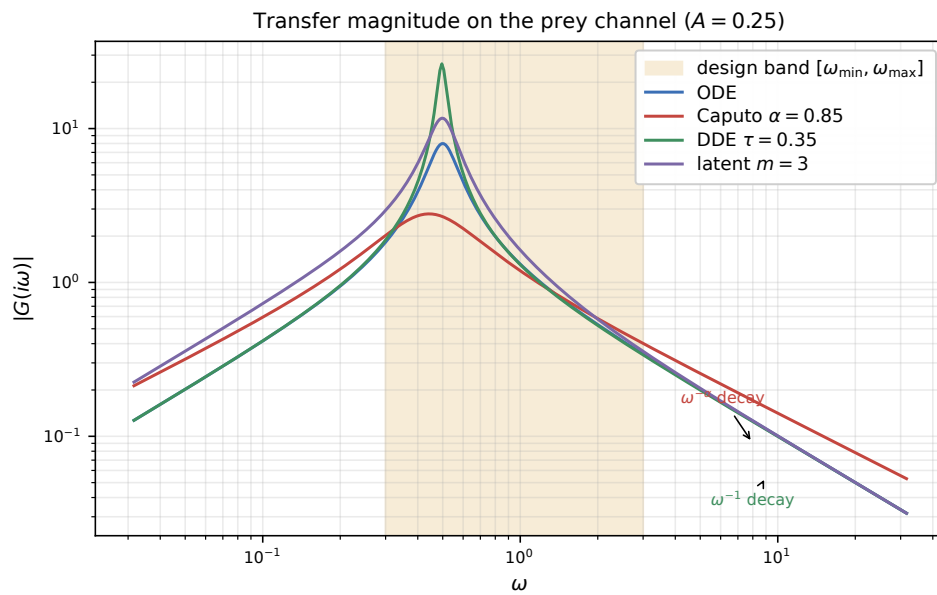


Figure 6. $|G(i\omega)|$ on the prey channel at $A = 0.25$ for ODE, Caputo ($\alpha = 0.85$), DDE ($\tau = 0.35$), and latent ($m = 3$). The Caputo curve separates over the design band and approaches its $O(\omega^{-\alpha})$ decay.

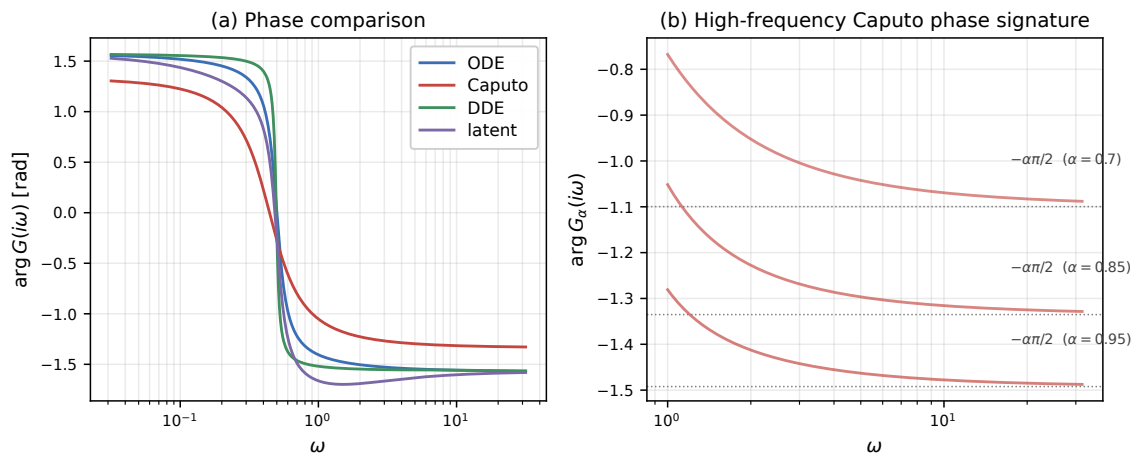


Figure 7. Phase on the prey channel. Left: the four mechanisms across frequency. Right: convergence of the Caputo phase to the asymptote $-\alpha\pi/2$, independent of scale.

Table 5. Certified finite-state approximation of the fractional prey response and induced testing lower bounds at $\alpha = 0.85$, $T = 12$, pulse $\|u\|_2 \leq 0.120$, and $\sigma = 0.10$. The exact bound is $\Phi(-d/2)$ with $d = \|\Delta\mu\|_2/\sigma$, valid because the two hypotheses share the covariance $\sigma^2 I$; Pinsker is reported for comparison and is uniformly looser.

m	$A = 0.25$			$A = 0.30$		
	$\hat{E}_m^{\text{state}}$	P_e^* (exact)	P_e^* (Pinsker)	$\hat{E}_m^{\text{state}}$	P_e^* (exact)	P_e^* (Pinsker)
4	0.5334	0.3745	0.340	0.4706	0.3888	0.359
8	0.1254	0.4700	0.462	0.0898	0.4785	0.473
16	0.0273	0.4935	0.492	0.0204	0.4951	0.494
32	0.0069	0.4983	0.498	0.0066	0.4984	0.498

0.4996 at $m = 64$ and 0.49999 at $m = 128$: at attainable latent orders the finite-state rival is indistinguishable from the fractional model under this protocol, and no experiment restricted to the declared budget can separate them.

Table 6. Prey-response L^1 error of an explicit positive-weight exponential-sum surrogate (not the certified infimum, hence an upper bound on $\widehat{E}_m^{\text{state}}$) and the induced exact testing floor, $\alpha = 0.85$, $A = 0.25$, $T = 12$, $\|u\|_2 \leq 0.120$, $\sigma = 0.10$.

m	$\ g_\alpha - g_m\ _{L^1(0,T)}$	quadrature error	P_e^* (exact)
4	$1.29 \cdot 10^0$	$2.1 \cdot 10^{-7}$	0.2202
8	$4.98 \cdot 10^{-1}$	$1.0 \cdot 10^{-7}$	0.3827
16	$1.30 \cdot 10^{-1}$	$7.6 \cdot 10^{-9}$	0.4689
32	$2.31 \cdot 10^{-2}$	$3.0 \cdot 10^{-8}$	0.4945
64	$1.58 \cdot 10^{-3}$	$2.1 \cdot 10^{-8}$	0.4996
128	$4.83 \cdot 10^{-5}$	$6.3 \cdot 10^{-9}$	0.49999

The table is one-sided. A large lower bound on P_e^* proves that the specified experiment cannot reliably separate g_α from at least one admitted rival g_m ; a small bound proves nothing in the converse direction, so such cells are inconclusive rather than certified-easy. Figure 8 applies this reading across (α, A, m) : cells outside the focal Matignon-stable regime ($\alpha > \alpha^*(A)$) are hatched and excluded from the headline interpretation.

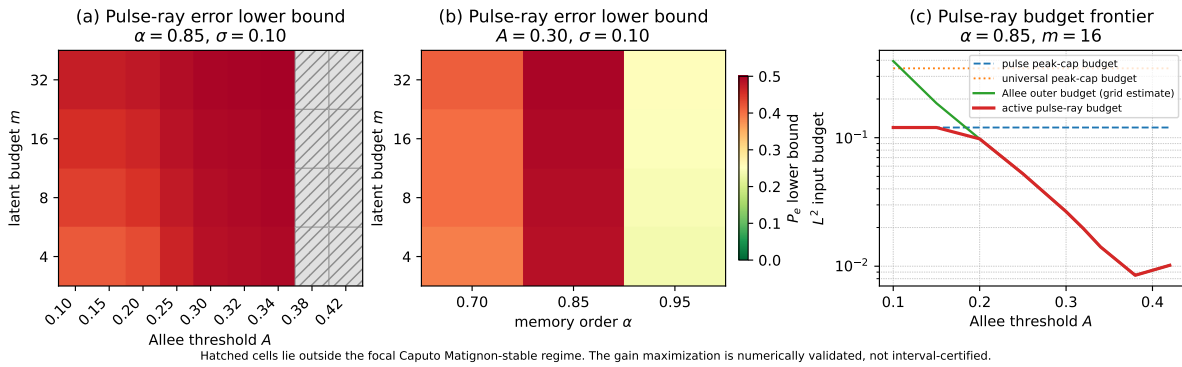


Figure 8. Protocol-restricted discrimination atlas over (α, A, m) : lower bound on the model-selection error under the declared pulse protocol. The bound increases with latent budget m and toward weakly separated regimes; hatched cells violate the Matignon condition $\alpha < \alpha^*(A)$ and are excluded from stable-regime conclusions.

6. Fractional-Delay Model: Dedicated Simulation Study

Sections 3–5 treated fractional memory and explicit delay as competing hypotheses. This section studies the mechanism carrying both, via the combined model (2.4). Its boundaries coincide with the two mechanisms already characterized: $\tau = 0$ is the pure Caputo model and $\alpha = 1$ is the DDE benchmark, with both limits verified to the tolerances of Table 3. The interior of $(\alpha, \tau) \in [0.70, 1] \times [0, 0.5]$ then measures whether the two effects reinforce, cancel, or generate responses outside the reach of either pure explanation.

The scope is deliberately limited: no general theory of fractional delay equations is attempted. The study is a stress test of the discrimination framework under identical conditions to the rest of the paper — same ecology, same prey channel, same budget $\|u\|_\infty \leq 0.10$, same observables — with only (α, τ) varying. What it exposes is invisible to pairwise comparisons: when a system carries a hereditary response *and* a physiological lag, a binary “fractional versus delay” classification is misspecified even if each limiting model fits part of the data.

6.1. Transient trajectories

Figure 9 fixes $A = 0.25$, $\alpha = 0.85$, applies the identical prey pulse, and varies $\tau \in \{0, 0.15, 0.35, 0.50\}$. The prey trajectory changes little; the predator response shifts visibly with τ and its late transient becomes increasingly delay-sensitive. All four trajectories satisfy $x(t) > A$ throughout, with minimum margin ≥ 0.356 (Table 7).

6.2. Fractional-order/delay parameter plane

Figure 10 maps the 7×6 grid $\alpha \in \{0.70, \dots, 1.00\}$, $\tau \in \{0, \dots, 0.50\}$. The edge $\tau = 0$ has zero distance to the pure Caputo model and the edge $\alpha = 1$ zero distance to the DDE, by construction; the quantity of interest is the interior

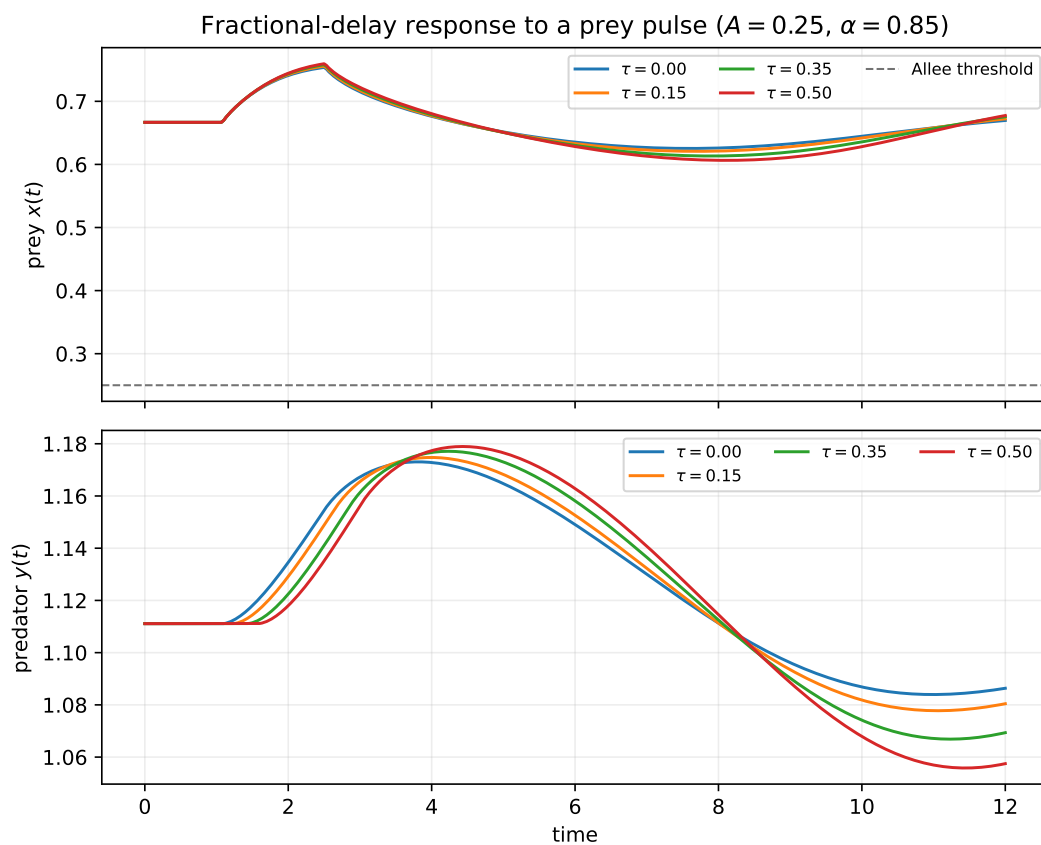


Figure 9. Combined fractional-delay response to the same prey pulse at $A = 0.25$ and $\alpha = 0.85$. Increasing delay shifts the predator response and reshapes the late transient while the fractional-memory relaxation pattern persists.

escape rate from both. The right panel adds the safety dimension: a delay that leaves the margin nearly untouched at $\alpha = 0.75$ (from 0.3917 to 0.3850 over $\tau \in [0.1, 0.5]$) removes 21% of it at $\alpha = 0.95$ (from 0.3345 to 0.2629).

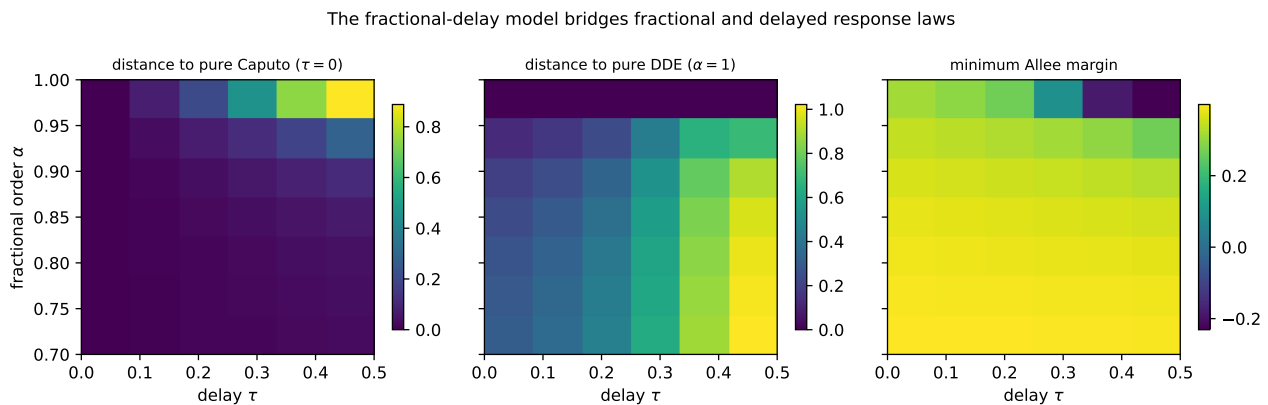


Figure 10. Fractional-delay bridge over (α, τ) . Left: trajectory distance to the pure Caputo limit. Center: distance to the pure DDE limit. Right: minimum Allee margin under the same pulse. Parts of the interior are poorly represented by either limit alone.

Table 7 lists nine representative cells. For each fixed α , d_{Caputo} increases monotonically in τ while d_{DDE} remains large (≥ 0.16 in every displayed cell). The steepest row is $\alpha = 0.95$: d_{Caputo} grows by a factor of 9.4 (from 0.0299 at $\tau = 0.10$ to 0.2805 at $\tau = 0.50$) while the minimum margin falls from 0.3345 to 0.2629. In this corner of parameter space, added delay increases identifiability and ecological risk simultaneously.

Table 7. Representative cells from the fractional-delay bridge at $A = 0.25$ under the pulse experiment. Distances are trajectory L^2 distances to the corresponding limiting models.

α	τ	distance to Caputo	distance to DDE	min. $[x(t) - A]$
0.75	0.10	0.0063	0.3426	0.3917
0.75	0.30	0.0182	0.6145	0.3887
0.75	0.50	0.0315	1.0086	0.3850
0.85	0.10	0.0103	0.2900	0.3724
0.85	0.30	0.0334	0.5683	0.3655
0.85	0.50	0.0625	0.9606	0.3566
0.95	0.10	0.0299	0.1645	0.3345
0.95	0.30	0.1142	0.4330	0.3123
0.95	0.50	0.2805	0.6875	0.2629

6.3. Which input exposes the combined mechanism?

Each input is scored over the nine cells $\alpha \in \{0.75, 0.85, 0.95\} \times \tau \in \{0.15, 0.30, 0.45\}$ by the mean of $\min(d_{\text{Caputo}}, d_{\text{DDE}})$; the min penalizes inputs that separate the combined response from only one limit. Results (Figure 11, Table 8): multisine attains the highest score (0.1332) with crossing fraction 1/3; PRBS (0.0909), sinusoid (0.0894), and chirp (0.0861) cross in all nine cells; pulse (0.0627) and multiscale (0.0338) never cross but discriminate weakly. The safety–informativeness tension of Section 8 reappears unchanged in the two-memory model.

Table 8. Fractional-delay design stress test at $A = 0.25$. The score averages the minimum distance to the Caputo and DDE limits over $\alpha \in \{0.75, 0.85, 0.95\}$ and $\tau \in \{0.15, 0.30, 0.45\}$.

Input	mean bridge score	worst Allee margin	crossing fraction
multisine	0.1332	-0.2103	0.333
PRBS	0.0909	-0.2814	1.000
sinusoid	0.0894	-0.2777	1.000
chirp	0.0861	-0.2763	1.000
pulse	0.0627	0.2770	0.000
multiscale	0.0338	0.3171	0.000

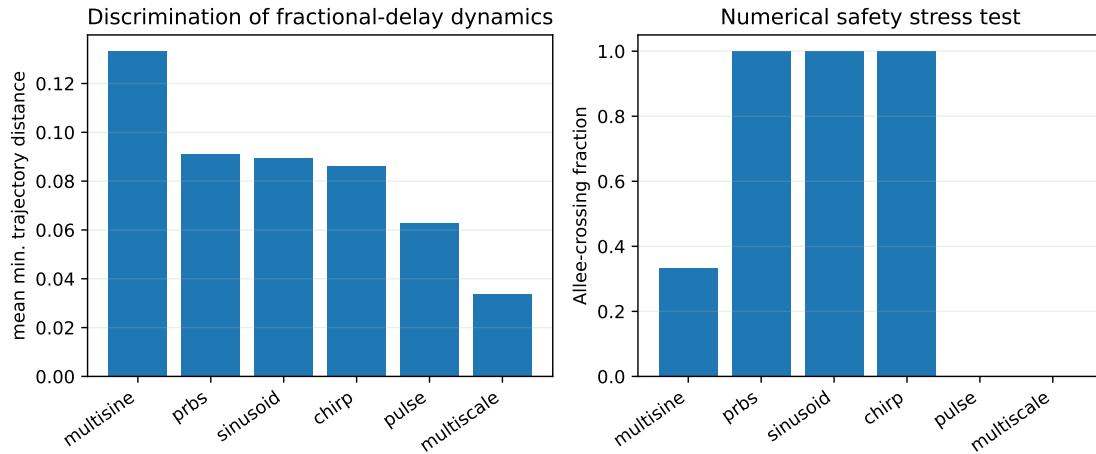


Figure 11. Input-design stress test for fractional-delay dynamics. Left: mean minimum trajectory distance from the two limiting explanations (pure Caputo and pure DDE). Right: numerical Allee-crossing fraction on the same (α, τ) grid. Broadband inputs expose the combined mechanism more strongly and are substantially less safe.

6.4. Delay sweep

Figure 12 sweeps $\tau \in [0, 0.55]$ at twelve points for $\alpha \in \{0.75, 0.85, 0.95\}$. In all three orders the predator peak time increases with τ , its amplitude grows moderately (by $< 1\%$), and the minimum Allee margin decreases — weakly at $\alpha = 0.75$, strongly at $\alpha = 0.95$. Delay is thus observable as a reproducible timing shift, not merely as a fitting parameter, and the shift carries a quantifiable safety cost.

Endpoints (Table 9): at $\alpha = 0.85$, the change $\tau : 0 \rightarrow 0.55$ moves the predator peak from $t = 3.814$ to $t = 4.500$ (+0.686) and the margin from 0.3759 to 0.3548 (−5.6%); at $\alpha = 0.95$ the same change moves the peak from $t = 4.157$ to $t = 4.886$ and the margin from 0.3434 to 0.2384 (−30.6%).

Table 9. Endpoint summary of the delay sweep under the pulse experiment. Values compare $\tau = 0$ with $\tau = 0.55$.

α	τ	predator peak time	predator peak	min. Allee margin
0.75	0	3.343	1.1634	0.3934
0.75	0.55	4.029	1.1681	0.3843
0.85	0	3.814	1.1724	0.3759
0.85	0.55	4.500	1.1788	0.3548
0.95	0	4.157	1.1838	0.3434
0.95	0.55	4.886	1.1925	0.2384

The sampling-design implication is direct. Delay is expressed in the timing of the predator peak (at $t \approx 3.3$ –4.9 across the tested grid); fractional memory in the relaxation tail beyond it. A window restricted to the initial prey perturbation misses both signatures: it ends before the delayed predator response and far before the tail. Dense sampling around the predicted peak, extended through the tail, captures both — the recommendation quantified in Section 9.

7. Experiment Design and Safety

The design theory itself is classical in outline; we state just enough here to motivate the waveforms, and refer the full development with proofs to the Supplementary Material. For two Gaussian linearized models with common covariance, the pairwise information carried by an input u takes the quadratic form

$$D_{ij}(u) = \frac{1}{2} \langle u, Q_{ij} u \rangle,$$

so under the energy constraint $\|u\|^2 \leq E$ the optimum is the leading eigenfunction of Q_{ij} , and the robust frequency-domain problem admits an optimum with finite frequency support (both proved in the Supplementary Material). These two facts justify restricting the search to the six implementable waveform families of Figure 13 instead of an infinite-dimensional input space.

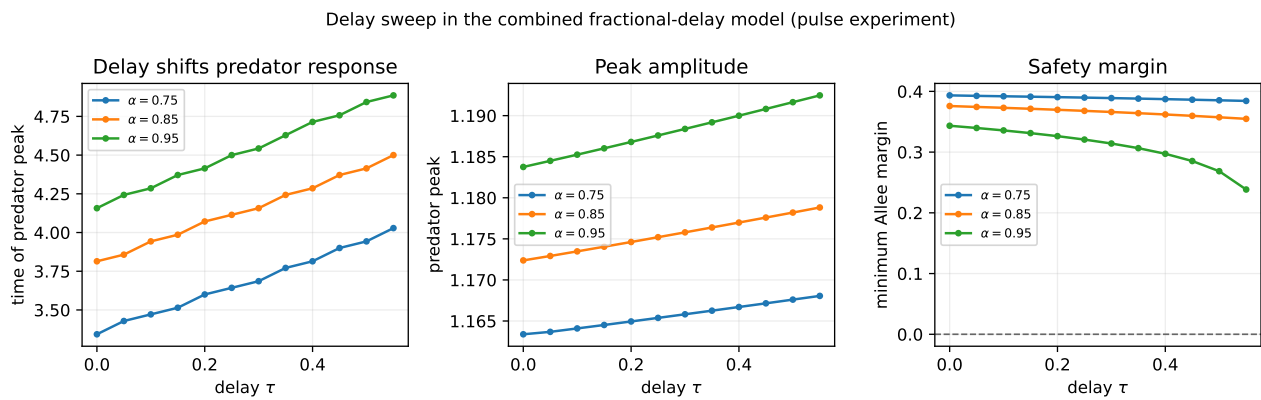


Figure 12. Delay sweep under the pulse experiment. The predator peak arrives later as τ grows, its amplitude increases moderately, and the numerical Allee margin shrinks — consistently across the three displayed fractional orders.

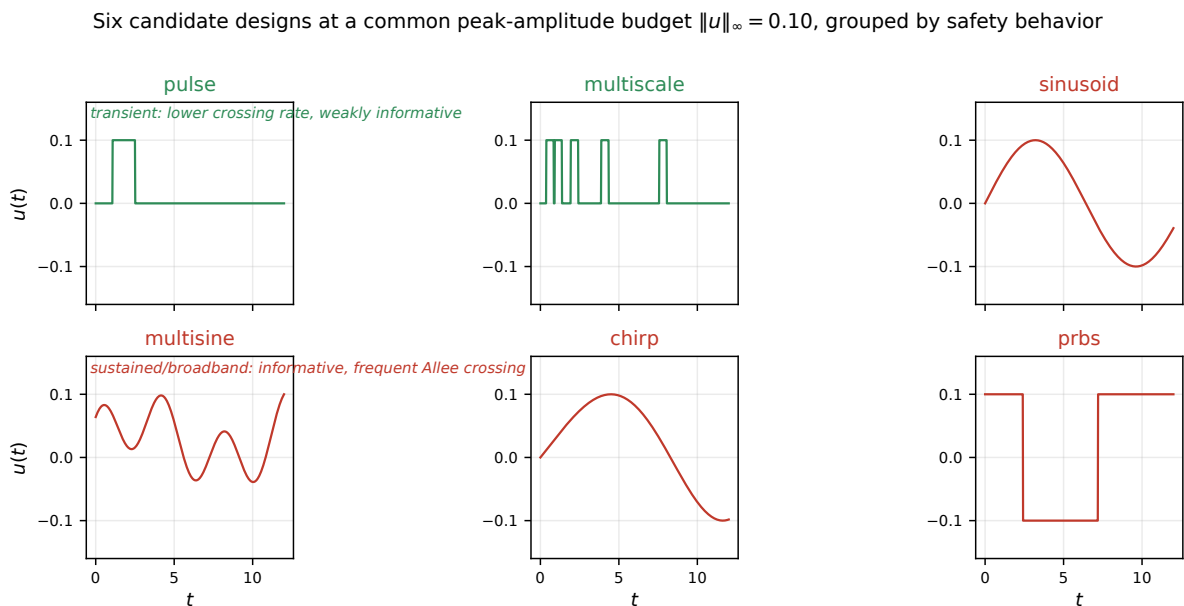


Figure 13. Six candidate perturbations at the common peak-amplitude budget. Transient and broadband designs interrogate the memory law in different ways — and behave very differently with respect to safety once the nonlinear simulations run.

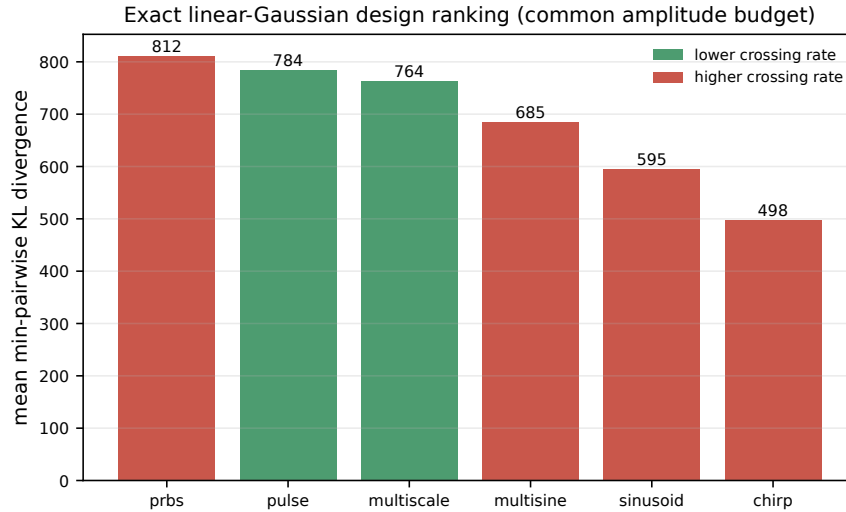


Figure 14. Linear-Gaussian design ranking by mean minimum pairwise KL divergence. Nonlinear state safety is invisible at this level, so the ranking serves as an information benchmark, not as a final experimental recommendation.

7.1. Strong-Allee safety

The one-sided Allee constraint requires $x(t) > A + \eta$ for all $t \in [0, T]$. For the linearized system, a sufficient pulse-amplitude envelope follows from the prey impulse gain; for the nonlinear system, the strict inward-pointing rectangle theorem and its four face inequalities are given in the Supplementary Material. One structural point belongs in the main text: state safety alone does *not* bound input energy. A stable strictly proper channel attenuates high frequencies as $O(\omega^{-1})$ or faster, so an oscillatory input of arbitrarily large $\|u\|_2$ can leave the state nearly unmoved. Every quantitative testing bound in this paper is therefore tied to an explicit actuator or protocol class — here, the declared peak-amplitude pulse budget $\|u\|_\infty \leq 0.10$, $\|u\|_2 \leq 0.120$.

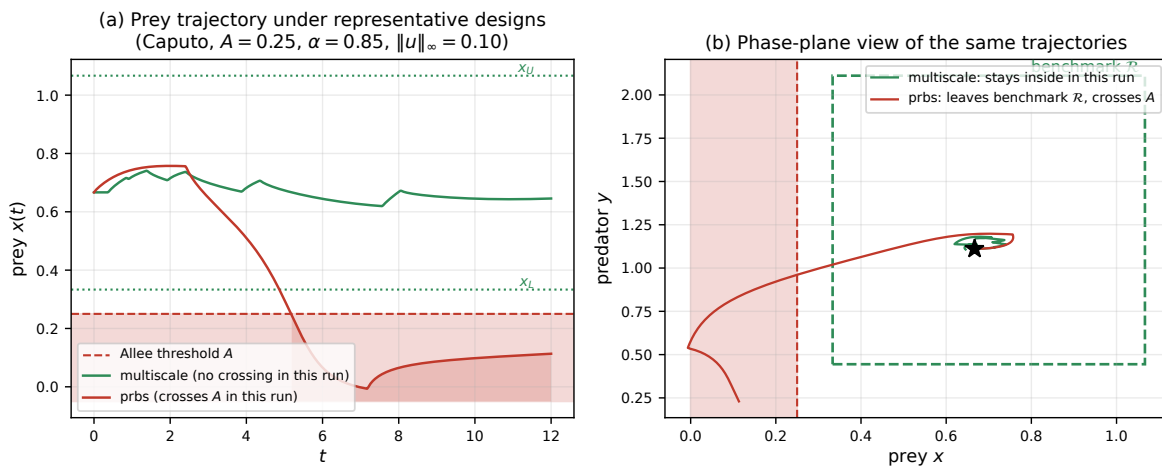


Figure 15. Representative nonlinear safety geometry. In the run shown, a multiscale transient stays above the Allee threshold while PRBS crosses it. The rectangle drawn is a benchmark diagnostic region; rigorous positive invariance requires the strict face conditions stated in the Supplementary Material.

8. Large-Scale Model-Discrimination Benchmark

The benchmark compares ODE, Caputo, retarded DDE, and latent-3 mechanisms at matched operating conditions. The primary stable-regime task uses $A \in \{0.20, 0.25\}$; a separate task at $A > 2/7$ addresses the easier stability-verdict question. The decision rule is BIC throughout — an information criterion, not Bayesian model evidence, and it is reported as such.

8.1. Benchmark design and decision rule

The benchmark is not organized around any single synthetic trajectory. It asks whether the qualitative design conclusions survive joint variation of memory strength (α), observation noise (three SNR levels), observation channel (three), and waveform (six families). For each generated trajectory, all four candidates are fitted under one observation protocol and compared by BIC; the object under evaluation is a reproducible decision rule.

Two factorial layers are kept separate. The nonlinear stable-regime benchmark has 1512 cells $\times$ 200 stochastic replicates (302,400 trajectories, zero solver divergences); a linear-Gaussian factorial of 1458 cells serves only as an information-ranking reference (Figure 14). The layers can disagree: the linear ranking measures local operator separation, while the nonlinear layer sees the Allee geometry, and an input that tops the linear ranking is disqualified once its trajectories cross $x = A$ (Table 11).

Macro-averaged accuracy in the primary four-class stable task is 0.537 versus chance 0.25. By input family:

$$\begin{aligned} \text{PRBS (0.584)} &\approx \text{multisine (0.583)} > \text{sinusoid (0.534)}, \\ &> \text{chirp (0.507)} > \text{pulse (0.414)} > \text{multiscale (0.324)}. \end{aligned}$$

Difficulty is dominated by two axes (Table 10). As $\alpha \rightarrow 1$ the mechanisms converge and accuracy falls from 0.702 ($\alpha = 0.70$) to 0.312 ($\alpha = 0.95$); across noise, accuracy is 0.756/0.444/0.273 at high/medium/low SNR. The channel effect is smaller: both-species observation attains 0.530, prey-only 0.516 (the directly actuated channel remains structurally sufficient), predator-only 0.427.

Table 10. Stable-regime four-class BIC accuracy stratified by fractional order, SNR, and observation channel.

Axis	level	accuracy	Axis	level	accuracy
α	0.70	0.702	SNR	high	0.756
	0.85	0.557		medium	0.444
	0.90	0.467		low	0.273
	0.95	0.312	channel	prey	0.516
				predator	0.427
				both	0.530

Class-wise statistics refine the picture (Figure 16). ODE: recall 0.972, precision 0.227 — under weak excitation the subtler mechanisms are absorbed into the simplest class. Caputo: recall 0.524, precision 0.984 — a Caputo verdict, once issued, is nearly always correct. DDE: 0.410/0.731; latent-3: 0.242/0.848. The dominant failure mode is therefore conservative collapse toward the simpler explanation when the data underdetermine the memory law, not symmetric confusion among mechanisms.

8.2. The trade-off within classical waveform families

Numerical state safety is reported on its own axis, never folded into the information score. At the common budget $\|u\|_\infty \leq 0.10$: multiscale never crosses the threshold (rate 0.000) and discriminates worst (0.324); PRBS and sinusoid discriminate best (0.584, 0.534) and cross at rates 1.000; multisine matches PRBS in accuracy (0.583) at crossing rate 0.357.

Table 11 gives the frontier *within the six classical families of Section 7*. Section 8.4 shows that this frontier is a property of that parameterisation and not of the system. Multiscale retains a positive minimum margin of 0.292 at the lowest accuracy; the pulse is the practical compromise, crossing at rate 0.071 with accuracy 0.414 — below the broadband designs by 0.17.

Table 11. Nonlinear stable-benchmark trade-off at the common peak-amplitude budget. Safety values are numerical benchmark diagnostics, not invariant-set certificates.

Input	macro accuracy	Allee crossing rate	minimum Allee margin
PRBS	0.584	1.000	-0.3139
multisine	0.583	0.357	-0.2405
sinusoid	0.534	1.000	-0.3058
chirp	0.507	0.929	-0.3012
pulse	0.414	0.071	-0.1207
multiscale	0.324	0.000	0.2921

The table also fixes a methodological point: the linear-Gaussian ranking alone cannot select the final experiment. Linear theory scores an input by operator separation; the nonlinear ecology can answer the same input by leaving the

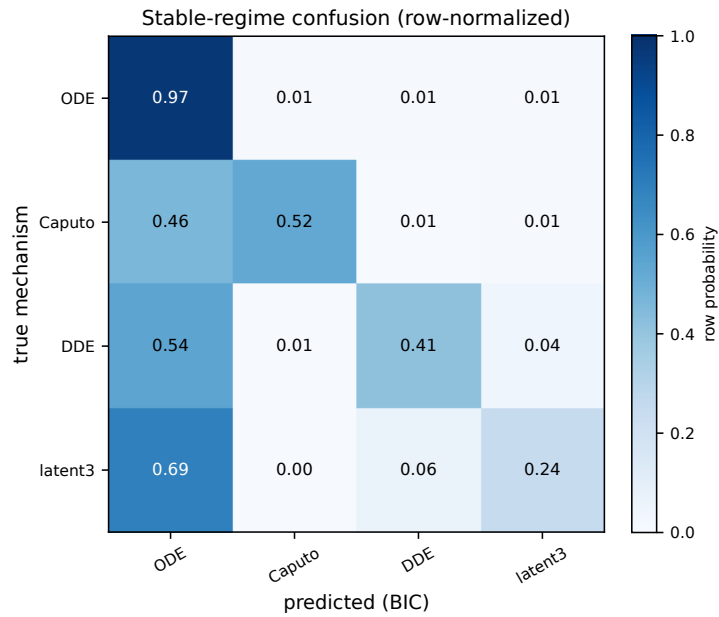


Figure 16. Row-normalized confusion matrix for the stable four-class task. Under weak excitation the simplest ODE mechanism absorbs the rest; Caputo predictions, when made, are highly precise.

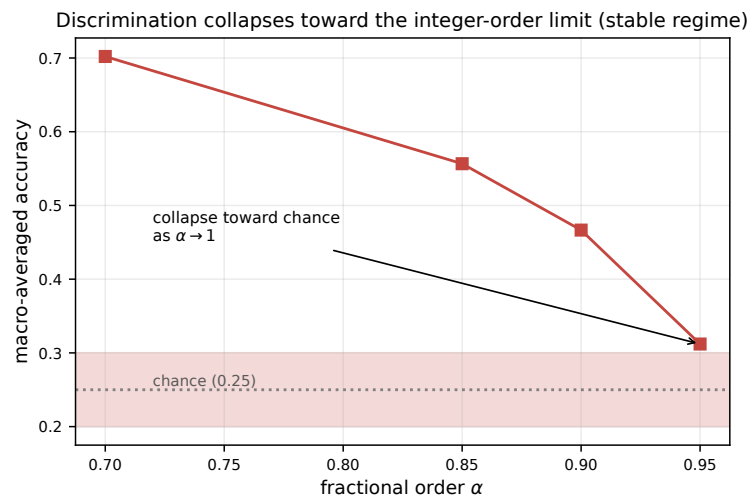


Figure 17. Model-selection accuracy versus fractional order. The closer α gets to the integer-order limit, the harder the memory mechanisms are to tell apart.

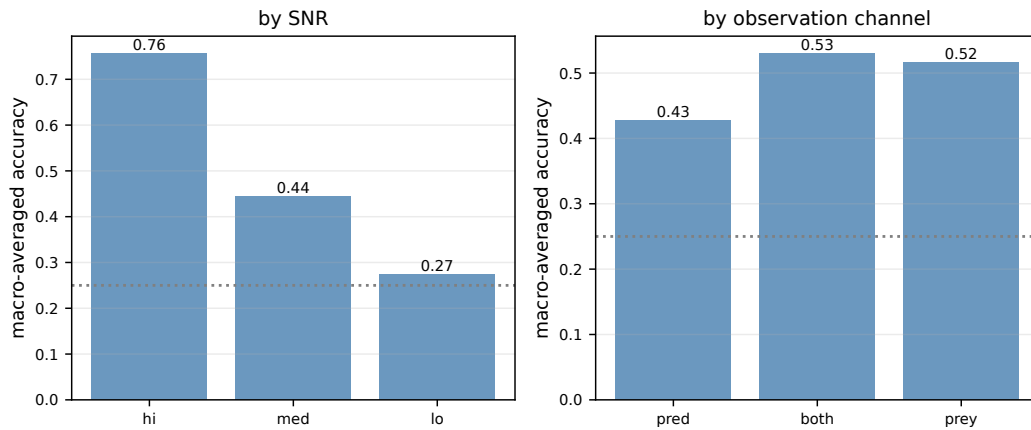


Figure 18. Accuracy stratified by SNR and observation channel. The benchmark is strongly noise-limited; observing both species gives the best average accuracy.

coexistence basin, after which no local identification experiment remains. Safety must be evaluated on the nonlinear state trajectory, not extrapolated from the linear information score.



Figure 19. Trade-off across the six classical waveform families. Broadband and sustained designs climb in discrimination but drift toward higher Allee-crossing rates; transient designs stay in the low-crossing, less-informative corner. The frontier shown is specific to these families: the piecewise-constant designs of Section 8.4 lie above and to the left of every point plotted here.

8.3. Amplitude is the wrong knob

Before accepting the frontier of Table 11 one must ask whether it is simply a symptom of excessive excitation. It is not. Repeating the entire benchmark at peak amplitudes 0.050 and 0.063 alongside the reference 0.100 (2268 cells in total, zero divergences) leaves the ordering of designs unchanged and does not restore safety: PRBS remains the most informative design at every amplitude while still crossing the threshold in 71.4% of cells at half amplitude (92.9% at 0.063, 100% at 0.100). Macro-accuracy falls monotonically with amplitude (0.457, 0.471, 0.537), so the low-amplitude runs are reported as linear-certificate diagnostics, not as a safe benchmark.

The realised response gain $\max_t |z(t) - z^*| / \|u\|_\infty$ makes the mechanism explicit. For the crossing designs this gain *increases* as the amplitude decreases, reaching 28.07 at amplitude 0.050 against the linear certificate $\Gamma_T = 5.78$, a violation by a factor of 4.86; for the two designs that never cross it stays between 1.47 and 2.06 and is essentially amplitude-independent, as a linear response must be. Once a trajectory leaves the Allee basin, $\max_t |z - z^*|$ is set by

the basin geometry rather than by the input, so the ratio scales like $1/\|u\|_\infty$. The linear certificate therefore cannot be used to justify safety for any crossing design at any of the three amplitudes, and shrinking the input does not convert an unsafe design into a safe one.

8.4. Waveform search: the frontier is a parameterisation artifact

If amplitude is not the knob, the waveform shape may be. We therefore searched for inputs that are safe *and* informative outside the six classical families, at the same peak-amplitude budget $\|u\|_\infty \leq 0.100$ and under the hard constraint that the minimum Allee margin over all four candidate models be at least $\delta = 0.05$. The search covers piecewise-constant inputs with 4, 6 and 8 levels, free multisines with 3–6 harmonics, and pulse trains with 2–4 pulses, scored by the minimax pairwise separation $J(u) = \min_{a \neq b} \|\mu_a(u) - \mu_b(u)\|/\sigma$ over the four classes; 163 840 candidates were evaluated by Sobol sampling with Nelder–Mead refinement of the per-family best, and 87 601 of them (53.5%) satisfy the safety constraint. Re-scored under the same objective, exactly one of the six classical designs is safe at this amplitude (multiscale, $J = 1.45$) and it is the least informative of the six; the best safe candidate found reaches $J = 6.23$, a factor of 4.3 higher, and every one of the twelve best safe candidates is piecewise constant — a class the six families do not contain.

Validating the six leading candidates with the same BIC classifier, the same 56-cell grid and the same replicate seeds as Table 11 confirms the surrogate (Table 12). All six are safe, with zero crossings and certified margins between +0.067 and +0.176, and five of the six outrank the best classical design of *any* kind, safe or not. The leading design attains macro-accuracy 0.803 against 0.514 for the only safe classical design and 0.763 for multisine, which crosses in 35.7% of cells. The gain is concentrated exactly where the safe classical design fails: Caputo recall rises from 0.404 to 0.911 and DDE recall from 0.445 to 0.853.

Table 12. Waveform search validated with the benchmark classifier at $\|u\|_\infty \leq 0.100$: 56 cells $\times$ 100 replicates per design, identical seeds, no divergences. pwck denotes a k -level piecewise-constant input on equal sub-intervals.

design	family	macro	micro	crossing rate	min. margin	Caputo	DDE
S0	pwck6	0.803	0.787	0.000	+0.092	0.911	0.853
S3	pwck6	0.783	0.753	0.000	+0.122	0.910	0.927
S2	pwck6	0.783	0.765	0.000	+0.096	0.875	0.816
S1	pwck6	0.780	0.744	0.000	+0.109	0.838	0.897
S5	pwck8	0.769	0.727	0.000	+0.176	0.844	0.926
multisine	classical	0.763	0.725	0.357	−0.240	0.844	0.902
S4	pwck6	0.750	0.720	0.000	+0.067	0.766	0.743
prbs	classical	0.711	0.713	1.000	−0.314	0.802	0.521
sinusoid	classical	0.685	0.687	1.000	−0.306	0.844	0.549
chirp	classical	0.683	0.684	0.929	−0.301	0.832	0.547
pulse	classical	0.621	0.552	0.071	−0.121	0.528	0.647
multiscale	classical	0.514	0.444	0.000	+0.292	0.404	0.445

Two qualifications are required. The search establishes *existence* of safe designs above the classical frontier, not global optimality: it is a Sobol-plus-local-refinement search, and no certified global bound over the piecewise-constant class is claimed. And the constraint is active — the refined optimum sits at the declared margin $\delta = 0.05$, while multiscale sits at 0.292 — so a stricter margin buys a lower objective and the trade-off is relocated rather than abolished. What does not survive is the reading of Table 11 as a property of the ecology: the accessible frontier lies well above it, and the residual obstruction is the latent one, since even the leading design recovers only 0.475 recall against the finite-latent rival.

9. Prospective Experimental Interpretation

Laboratory predator–prey microcosms provide the controlled perturbation setting the simulations presume [2, 7, 22], and the numerical results translate into specific protocol constraints. A first experiment should operate in a stable strong-Allee regime ($A \leq 2/7$), apply a bounded prey perturbation ($\|u\|_\infty \leq 0.10$ in normalized units), observe both populations where instrumentation allows (accuracy 0.530 versus 0.516 prey-only; Table 10), and postpone broadband excitation until a transient probe has established the available margin.

The fractional-delay study adds one operational diagnostic. Delay moves the predator peak time (by up to +0.73 time units over $\tau \in [0, 0.55]$; Table 9); fractional order reshapes the long transient and phase. A scheme sampling densely around the predicted peak ($t \approx 3.3$ – 4.9 on the tested grid) and recording through the relaxation tail resolves

both signatures; uniform sparse sampling can miss both.

A Bayesian sequential layer is specified in the Supplementary Material and deliberately not executed; the evidence in the main text is limited to deterministic solvers, interval enclosures, and BIC benchmarks, all rerunnable from the released scripts.

A two-stage prospective protocol

The observed frontier (Tables 11 and 8) argues against applying the most informative perturbation first. Stage 1 uses a transient input with low observed crossing rate (pulse, 0.071; multiscale, 0.000) to estimate the local response scale and verify a comfortable margin above A . Stage 2 is conditional: multisine-type excitation (accuracy 0.583, crossing rate 0.357) is applied only if Stage 1 leaves adequate margin. No Bayes-optimality is claimed for this sequencing; it is a direct reading of the frontier, summarized in Table 13.

Table 13. Prospective microcosm workflow suggested by the simulation results.

Stage	Perturbation	Primary measurements	Decision / purpose
0	no perturbation	baseline prey/predator variability	estimate noise and confirm operating point
1	pulse or multiscale	dense early prey observations; predator peak timing; minimum prey level	verify local response and available safety margin
2	multisine if margin permits	both species over the full transient and relaxation tail	maximize separation of memory mechanisms
3	optional repeat / altered horizon	same channels, replicated under matched conditions	test robustness of the selected mechanism and estimate variability

Three measurement priorities follow from the simulations. (i) The actuated prey channel carries the strongest early signal: sample densely immediately after the perturbation. (ii) The delayed pathway is expressed in predator-peak timing: concentrate observations near the anticipated peak rather than uniformly. (iii) Fractional memory dominates the slow relaxation tail: extend the horizon well past the peak. The resulting design is nonuniform — dense around the forcing and the peak, sparser but longer through relaxation.

On units: the simulations are dimensionless with a normalized budget. A physical implementation must map state and input to organism counts, biomass, or resource-manipulation units and must fix an operational margin above the *empirical* Allee threshold. This calibration is application-specific and is not inferred from simulation here.

10. Discussion

Three conclusions are supported consistently by the analytical results of Section 3 and the numerical layers of Sections 5–8.

First, *structural identifiability and practical discrimination are different questions*. The Caputo transfer carries an exact $O(\omega^{-\alpha})$ signature (Theorem 3.1); finite latent models approximate the same response to any tolerance on $[0, T]$ (Theorem 3.2); and the testing bounds quantify the gap — $P_e^* \geq 0.498$ at $m = 32$ under the pulse protocol (Table 5). A flexible latent rival therefore remains problematic long after the model classes are proved analytically distinct.

Second, *delay and fractional memory should not be treated as mutually exclusive*. The simulations of Section 6 trace a continuous bridge between the Caputo limit ($\tau = 0$) and the DDE limit ($\alpha = 1$): delay governs predator-response timing, fractional order the relaxation structure, and interior cells sit far from both limits simultaneously ($d_{\text{Caputo}} = 0.28$, $d_{\text{DDE}} = 0.69$ at $(\alpha, \tau) = (0.95, 0.50)$; Table 7). Fitting the two endpoint models separately cannot reveal this; the composite stress test can.

Third, *within classical waveform families the most informative perturbation is not the safest*, by a wide margin: the accuracy-leading inputs cross the Allee threshold at rates 0.357–1.000 in the four-class benchmark (Table 11) and in 3/9 to 9/9 cells of the fractional-delay grid (Table 8), two independent measurements of the same effect. Crossing A is not a numerical artifact: it changes the basin structure, and the procedure ceases to be a local identification experiment at that point.

Fourth, and this qualifies the third conclusion decisively, *that trade-off belongs to the parameterisation and not to the ecology*. Neither reducing amplitude (Section 8.3) nor accepting the classical frontier is necessary: searching piecewise-constant inputs at the same amplitude budget under a hard margin constraint produces designs that are simultaneously safe and more informative than every classical design, safe or not — macro-accuracy 0.803 with zero crossings against 0.514 for the only safe classical family and 0.763 for the best classical design overall (Table 12). The frontier of Table 11 is therefore a statement about six waveform families, and reporting it as a property of safe

ecological identification would be an error. What remains binding is the latent-complexity obstruction of Theorem 3.2 and Tables 5–6: even the leading safe design recovers only 0.475 recall against the finite-latent rival, and the induced testing floor reaches 0.49999 by $m = 128$. The operative constraint on identifying ecological memory is the declared rival-complexity budget, not the safety margin.

Limitations. Theorem 3.4 is linearized and specific to the prey channel; a full-state, hierarchy-wide safety closure remains open. The waveform search of Section 8.4 establishes existence, not global optimality: no certified global bound over the piecewise-constant class is claimed, the search was run at a single amplitude and a single α , and its safety constraint is active at the declared margin. Safety of the leading designs is certified a posteriori by mesh refinement together with a rigorous inter-node bound from the Caputo modulus of continuity, which decides every trajectory tested but is not an interval enclosure of the solver error. The fractional-delay study is numerical only, with constant equilibrium history, one delayed trophic pathway, and the targeted grids of Table 4. The benchmark decides by BIC, not Bayesian evidence; adaptive design is specified (Supplementary Material) but not executed. This ledger separates proved results from computational evidence.

Relation to the previously submitted companion paper. The shared material is the strong-Allee backbone with its locked parameterization, from Ref. [12]; nothing else is shared. Ref. [12] concerns certification: computer-assisted verification of Caputo equilibria, the Matignon boundary $\alpha^*(A)$, extinction/boundedness, and the diagnostic failure of a forward-Euler surrogate. The present work concerns discrimination. Theorems 3.1, 3.2, 3.3, and 3.4, the six-family design study, the 1512-cell benchmark, and the fractional-delay simulations appear nowhere in Ref. [12]. Each inherited stability fact enters only to select an admissible operating regime, and is attributed where it enters.

What is genuinely added by the present paper. Backbone reuse is an experimental-design decision. With the ecology frozen, differences between simulated observations are attributable to the memory law or to the experiment, never to an altered population model. The paper starts where [12] ends: a certified operating point leaves open whether an observed transient identifies a fractional mechanism uniquely, whether a delayed or latent model can reproduce it (Theorem 3.2 says yes, at sufficient m), and which perturbations separate the alternatives within the safe regime. These questions require the mathematics of Section 3 and the computations of Sections 4–8, none of which appears in [12].

The figure sets reflect the same division: the centerpiece of [12] is a certified (A, α) stability atlas, whereas Figures 6–19 here concern response separation, approximation error, waveform design, BIC confusion, fractional-delay dynamics, and the safety–informativeness frontier. Only the model schematic (Figure 2) orients the reader in the common ecology; no certificate atlas or certificate count from [12] is used as evidence.

Relation to broader prior work. Safe information limits and safe active model discrimination exist as general theories [15, 19]; no novelty is claimed for safe experimental design per se. The contribution is the specific quantitative interaction: finite-horizon fractional approximation (Theorem 3.2), bounded latent complexity, strong-Allee geometry, and the combined fractional-delay stress test in a single frame.

Implications for experimental identification. Model selection here is a *controlled intervention problem*, not passive curve fitting. Under weak excitation, BIC collapses to the ODE explanation (recall 0.972, precision 0.227); stronger excitation separates the mechanisms and simultaneously raises the crossing rate. The fractional-delay study adds a distinct warning: both endpoint models can be individually identifiable while an interior composite occupies a response region neither represents (Table 7), so designs should test composite alternatives, not only the endpoints.

The operative objective is not “maximize information” but maximize information *subject to* a trajectory-level safety margin and a declared rival-complexity budget — and Section 8.4 shows that this constrained problem should actually be solved rather than approximated by choosing among textbook waveforms, since the loss from not solving it is 0.29 in macro-accuracy. The second constraint is forced by Theorem 3.2: an unrestricted latent hierarchy reproduces fractional memory on any finite horizon, so the allowed latent dimension m , delay range, forcing bandwidth, and observation horizon T belong in the scientific specification of the experiment, alongside organism and protocol.

11. Conclusion

The question posed at the outset — what would it take, experimentally, to establish that an ecological system carries fractional memory — receives a quantitative and partly negative answer. Four analytical results in the main text (structural separation, finite-horizon latent approximation, the complexity-dependent testing obstruction, and the interval-certified prey-response approximation of Theorem 3.4) delimit what is provable; the numerical layers (frequency diagnostics, the fractional-delay study of Section 6, six input families with safety metrics, and the 1512-cell benchmark) measure what is achievable.

The measured answer is consistent across layers. Fractional memory is structurally distinct, yet at latent budget $m = 32$ the certified surrogate is within 0.002 of chance-level indistinguishability under the declared pulse protocol; excitation strong enough to separate the mechanisms (macro-accuracy up to 0.584) crosses the Allee threshold in up to 100% of stable cells. Before the next experiment, the productive question is therefore not *which model fits best* but *which safe perturbation makes the competing memory mechanisms observably different*.

Author contributions

The author contributed to the conceptualization, methodology, formal analysis, numerical investigation, visualization, and preparation of the manuscript.

Use of AI tools declaration

AI-assisted tools were used during manuscript preparation for language editing, LaTeX formatting, code and documentation assistance, and structural organization. Responsibility for the scientific content and the final submitted version remains with the author.

Conflict of interest

The author declares no conflict of interest.

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